

Group number: 40

Names of the members of the group: Francesco Binni, Shari Eggeringhaus, Benedikt Josef Erdt, Anna-Lina Matz, Luca Siegfried Müller, Marwin Paulick, Cesare Tomasi, Tino Emil Felix Weiss

Assignment 1 – Case Analyses

Part 1: Fast track Joint Replacement Surgery

Question 1

Restructuring projects, sometimes called “soft” projects, are mainly about changing how people work, think, and collaborate, rather than about producing a physical output. They usually need active change management, heavy stakeholder involvement, motivation building throughout the organization, and a project manager who genuinely understands the organizational culture.

You can see all of this in the fast track surgery case. The project restructured patient pathways, standardized evidence based practices, and pushed an entire orthopaedic department toward interdisciplinary collaboration. The “deliverable,” in the end, was really a new way of working.

Restructuring and construction projects do share some things: both operate under constraints, involve multiple stakeholders, and need careful planning. But construction produces something tangible and measurable, a bridge, a building, a road, and early decisions like procurement or contractor selection tend to lock in most of the project’s trajectory. Restructuring is different. The deliverable is behavioral and procedural; you can’t put a tape measure to it. Resistance to change is always lurking as a risk, and success hinges much more on communication, ownership, and gradual cultural alignment than on any technical specification.

Question 2

This project set out to drive several layered transformations at once. First, a clinical transformation: moving from traditional joint replacement care to a fast track model built around preoperative information, stress reduction, pain management, early mobilization, and nutrition. Second, a logistical transformation: standardizing patient pathways, cutting length of stay to under four days, and pushing throughput from 117 to 400 operations a year without adding operating theatre capacity. Third, an organizational transformation, since moving to the new mobility centre meant aligning surgeons, anaesthetists, nurses, physiotherapists, and technicians under one unified workflow. Fourth, a data and research transformation: building a quality indicator database and management tools to support transparency, clinical research, and evidence based improvement. And fifth, a cultural transformation, shifting from siloed professional autonomy toward joint decision making and interdisciplinary loyalty.

These changes mattered a lot. St. Olav's was Norway's second largest joint replacement centre, so changing how it operated didn't just affect internal logistics, it rippled into regional health delivery, waiting lists, and the hospital's research reputation. This wasn't a minor tweak to existing processes. It was a genuine re engineering of how care got delivered.

Question 3

This had to be run as a project because it ticked every box that defines one. It was temporary, with a clear start and end, aimed at a specific transformation under defined constraints. The objectives were concrete and measurable: specific operation volumes, waiting time limits, length of stay targets, satisfaction thresholds.

Running it as a project also had real advantages over treating it as routine operational work. As a strong matrix organization with expanded authority, the project could cut across departmental lines in a way ordinary line management never could. Handled as a regular task, it would have been swallowed by day to day clinical priorities. The steering committee, chaired by the CEO, gave the project a decision making body with enough power to enforce changes and settle conflicts between clinics. With fifteen representatives from the affected

entities, it created a legitimate space for interdisciplinary dialogue that normal operations just don't offer. And the project structure allowed for systematic documentation, follow up through the quality register, and clear accountability, all essential when you're transforming clinical pathways.

Part 2: Cost Optimization of a Product

Question 1

Complexity in projects shows up through organizational sophistication, a wide range of contributors, diversity in culture, geography, and discipline, and communication flows that get complicated fast. Once you have many units and individuals involved, especially spread across different locations and cultures, information stops flowing simply, and a linear plan just can't manage it anymore.

Bravo turned into exactly this kind of complex project, even though it didn't start out that way. Originally, Bravo was a small cost optimization effort inside a single department at one location. But once it morphed into Bravo plus, it became a huge, complex programme touching almost every department in the company across three continents and multiple time zones. It got tangled up in a prioritization matrix alongside thirty other concurrent projects. Functions like customer support, testing, and software development suddenly had to be pulled in too. That kind of multiplication, more stakeholders, more locations, more dependencies, more objectives, is pretty much the textbook definition of project complexity.

Question 2

The two projects had fundamentally different purposes. Bravo was about cost optimization: shrink the chip using improved process technology, meet the same specifications as Alpha 1, match or beat Alpha and Alpha 1 on performance, and work as a seamless drop in replacement. The goal was a full customer switch, retiring the older product lines, lower production costs, and capturing a narrow market window.

Bravo plus became something else entirely, a new product development project. Once marketing pushed for new functionality to make it compatible with the company's new platform, this stopped being a simple optimization job. It was building something new. That pulled in additional technical domains, required sign off from new departments, and killed the whole drop in replacement idea.

Management clearly didn't grasp this shift. They renamed the project but kept the same project manager, the same implicit timeline, and never renegotiated resources or governance. In effect, they treated a cross continental, multi departmental new product development effort as if it were still a small internal optimization task. That mismatch between what the project had actually become and the management framework still wrapped around it was a root cause of everything that went wrong afterward.

Question 3

Projects exist to create benefit for the project owner, by eliminating a problem, preventing one, or seizing a new opportunity. Bravo fits this perfectly. The original Alpha chip had been a big success, but Alpha 1 had a robustness problem that made roughly 40% of customers refuse to migrate. That forced the company into costly dual production of Alpha and Alpha 1, eating into profit. Bravo was launched to fix that problem and stop further customer loss to competitors, while also capturing the remaining market window before Moore's Law made the product obsolete.

In microelectronics, where market windows are narrow and price performance curves are unforgiving, cost optimization isn't just an engineering exercise, it's a matter of strategic survival. Bravo was tightly tied to the business goal of holding onto the client base, cutting operational costs, and defending competitive position. The real tragedy is that by mismanaging Bravo's evolution into Bravo plus, the company ended up undermining the very business goal it had set out to achieve.